

July 15, 2026

Editor-in-Chief
ACM Journal of Data and Information Quality (JDIQ)

Dear Editor,

We submit the anonymous manuscript “Data Quality for Derived Preference Graphs: Construction Sensitivity and Repair Outcomes in Multi-Ranker Retrieval” for consideration at the ACM Journal of Data and Information Quality.

The paper’s central contribution is methodological rather than algorithmic. It treats query-specific preference graphs as derived data artifacts whose properties depend on score normalization, vote construction, candidate pooling, conflict handling, and evaluation design before any repair method is applied. It then organizes that problem into a compact operational audit framework and evaluates the resulting construction sensitivity across four retrieval benchmarks.

The main empirical finding is that raw heterogeneous score margins create a scale-dominated graph, while normalized construction materially changes the graph being repaired. Repair remains structurally active, and under genuine $P > k$ evaluation it can change top- k membership, but no repaired-versus- unrepaired nDCG family in this study survives the primary Holm-corrected inference. Exact SCIP repair improves the structural objective without revealing a reliable retrieval gain that greedy repair missed.

We believe this framing fits JDIQ because it centers data quality in a derived analytical artifact, separates intrinsic graph quality from task-context utility, and converts that distinction into a reproducible audit procedure with practical guidance.

An anonymous supplementary ZIP accompanies the submission and includes the final supplement, canonical source code, aggregate intermediate tables, figure- and table-generation scripts, validation outputs, and frozen reproduction metadata.

Thank you for your consideration.

Sincerely,
Anonymous Author(s)